CCC–C–439d
May 1, 1964

SUPERSEDING
Int. Fed. Spec.
CCC–C–00439c (Army–GL)
January 29, 1964 and
Fed. Spec. CCC–C–439b
June 22, 1962

FEDERAL SPECIFICATION

CLOTH, BUNTING, COTTON, MERCERIZED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cotton bunting primarily intended for use in the manufacture of signal, semaphore, ensign (including U.S. flag), and other type flags.

1.2 Classification.

1.2.1 *Type.* Bunting covered by this specification shall be of one type only.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

P–S–617—Soap, Toilet, Hard, Soft or Sea Water.
CCC–T–191—Textile Test Methods.
PPP–P–51—Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cotton, Silks, and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.
Fed. Std. No. 102—Preservation, Packaging and Packing levels.
Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

Color Association of the United States, Inc.
Standard Color Card of America.

(Copies may be obtained from the Textile Color Card Association of the United States, Inc., 200 Madison Avenue, New York 16, N.Y.)

Federal Trade Commission Publication:
Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the finished cloth shall

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match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 *Yarn.* The yarn shall be two-ply in both warp and filling and shall be made from cotton which has been cleaned, carded, and drawn.

3.3 Color. The color shall be as specified (see 6.1 and 6.3). The use of sulfur dyes and dyes containing elementary sulfur or compounds capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed cloth shall show no more free or sulfide sulfur than the standard sample when tested as specified in 4.4. Unless otherwise specified when yellow bunting is required, and no standard sample is available, it shall match color number 23655 of Fed. Std. No. 595.

3.3.1 *Matching.* The color shall match the standard sample, when available, under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 *Colorfastness.* The dyed and finished cloth shall show colorfastness to crocking, light, water, and salt water and soap equal to or better than the standard sample when tested as specified in 4.4. When no standard sample is established, the dyed and finished cloth shall show fair colorfastness to crocking, and good colorfastness to light, water and salt water and soap when tested as specified in 4.4.

3.4 Weave. The weave shall be plain.

3.5 Width. The minimum width, exclusive of selvages, shall be 37½ inches for the Military services and 35¾ inches for Civil agencies (see 6.1).

3.6 Physical requirements. The finished cloth shall conform to the physical requirements shown in table I when tested as specified in 4.4.

TABLE I.—*Physical requirements*

Weight per square yard (minimum)	Yarns per inch (minimum)		Breaking strength (minimum)	
	Warp	Filling	Warp	Filling
Ounces			*Pounds*	*Pounds*
4.5	31	30	50	50

3.7 Finish. The cloth shall be fully mercerized. The pH shall be not less than 6.0 nor more than 9.0 when tested as specified in 4.4.

3.8 Nonfibrous materials. The total sizing, finishing, and other nonfibrous materials shall not exceed 2.0 percent when tested as specified in 4.4.

3.9 Length. Unless otherwise specified (see 6.1), the cloth shall be in continuous pieces of not less than 35 yards, and shall be put up in rolls or flat folded pieces as specified in PPP–P–51.

3.10 Fiber identification. Each piece shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.11 Workmanship. The cloth shall conform to the quality and grade of product established by this specification and the occurrence of defects shall not exceed the point level specified.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Examination of the end item. Examination of the end item shall be in accordance with the provisions of MIL–STD–105 and 4.2.1 through 4.2.4.

4.2.1 *Yard-by-yard examination.* The required yardage of each roll or piece shall be examined on the face side and all defects as defined in Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.1.1, except that spots, stains, and streaks shall be counted regardless of cause (i.e., not restricted to rust, oil, or grease) and only those slubs and knots which exceed the maximum acceptable limits shown for each shall be scored (see 4.2.1.2). Defects on the selvage shall not be scored unless the body of the cloth is affected. No yard shall be penalized more than four points. For purposes of this examination, a yard shall begin at the last scored defect. The sample unit for this examination shall be 1 linear yard. The sample size shall be 750 yards, regardless of lot size. The sample yardage shall be examined from 25 rolls or pieces taken from 25 containers. An approximately equal number of yards shall be examined in each roll or piece of the sample. The lot shall be unacceptable if the points per hundred square yards exceed 75.00 points. Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Sample size (yards)} \times \text{contracted width of material (inches)}} = \text{Points per 100 square yards}$$

4.2.1.1 *Demerit points.* Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	three points
For defects exceeding 9 inches in any dimension	four points

Note: The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth.
Objectionable odor.
Width less than specified.
Poor dye penetration, mottled, streaky or cloudy.

4.2.1.2 *Limits of slubs and knots.* Only slubs and knots which exceed the limits shown on figure 1 shall be scored.

4.2.2 *Examination for length.*

4.2.2.1 *Individual pieces.* During the yard-by-yard examination, each roll or piece shall be examined for length. Any length found to be less than the minimum specified or more than two yards below the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls or pieces in the sample are defective in respect to length.

4.2.2.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual lengths of rolls or pieces in the sample is less than the total of the lengths marked on the tickets.

4.2.3 *Examination for shade.* During the yard-by-yard examination, each roll or piece in the sample shall be examined for shade. Any piece in the sample off shade, shaded side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.4 *Examination for compliance with Textile Fiber Products Identification Act.* During the yard-by-yard examination, each roll or piece in the sample shall be examined for conformance to the Textile Fiber Products Identification Act. Each roll or piece not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.3 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers

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in the end item inspection lot. The inspection level shall be L–4 and the AQL shall be 4.0 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Weight	Gross or net weight exceeds requirements.

4.4 Tests. The methods of testing specified in CCC–T–191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit for test purposes shall be 1½ continuous yards full width of finished cloth. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

TABLE II.—*Test methods*

Characteristic	Requirement reference	Method number
Nonfibrous material	3.8	2611
Weight, ounces per square yard	Table I	5041
Yarns per inch	Table I	5050
Breaking strength	Table I	5100
Ply[1]	3.2.1	Visual
Weave[1]	3.4	Visual
pH	3.7	2811
Mercerization	3.7	Microscopic examination[1 4]
Sulfur content[1]	3.3	4.4.1
Colorfastness to:		
Crocking	3.3.2	5651
Light	3.3.2	5660[2]
Water	3.3.2	5630
Salt water and soap	3.3.2	5632[3]
Vat dye	3.3	(4)

[1] One determination per sample unit, and results reported as "pass or fail."

[2] In the event of dispute resulting from the test with method 5660, or as a result of suspected anomalous behavior of certain dye types or formulation, the contracting officer may authorize the exposure to natural light in accordance with method 5662.

[3] Method 5632 except that the soap shall meet the salt-water soap requirements of Fed. Spec. P-S-617.

[4] The contractor shall submit a certificate of compliance for these characteristics. The certificate shall be accompanied by actual test, inspection, or other verifiable quality data.

4.4.1 *Test for sulfur content.*

4.4.1.1 *Solution (acid stannous chloride).* Dissolve 100 grams of stannous chloride crystals in 100 ml. of hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water.

4.4.1.2 *Procedure.* One to 1.5 grams of finished cloth, cut into small pieces, shall be used for the test. Place the sample in a test tube of suitable size and cover with the stannous chloride solution. Place a filter paper which has

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been moistened with one drop of 5.0 percent lead acetate solution (made by dissolving 5 grams of reagent grade lead acetate to 100 ml. of distilled water; add a drop of glacial acetic acid to clear up if cloudy) over the mouth of the tube (punch needle hole in paper). Heat the test tube slowly over a low flame until the solution is boiling. Solution shall not be boiled for more than 10 to 15 seconds, as prolonged boiling will reduce the sensitivity of the test. A brown to black stain appearing on the paper indicates the presence of sulfur.

4.4.1.3 *Evaluation.* The stained filter paper shall then be compared with that obtained from the standard. If the stained area is equal to or lighter than the standard in the degree of staining, the sample shall be considered satisfactory. If the degree of staining by the sample is in excess of the standard, the sample shall be considered unsatisfactory.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1 and 6.2).

5.1.1 *Levels A and C.* The cloth, put up as specified, shall be packaged in accordance with the applicable requirements for class 3 fabrics of PPP–P–51.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1 and 6.2).

5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of PPP–P–51.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP–P–51.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Color required (see 3.3).
(*c*) Width required (see 3.5).
(*d*) When length of piece is other than specified (see 3.9).
(*e*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 In the preparation of contracts or orders it should be noted that various options, choices, and alternatives, as indicated in PPP–P–51 may be exercised in the preparation for delivery of this commodity.

6.3 Standard sample. For access to standard sample (see 3.1), address the procuring office issuing the invitation for bids. In case no standard sample is available for the United States Flag, the cable numbers contained in the Standard Color Card of America are as follows (see 2.2):

Red—70180
White—70001
Blue—70075

6.4 Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Cotton piece goods finished, not otherwise indexed by name:
Carload minimum weight—any quantity.

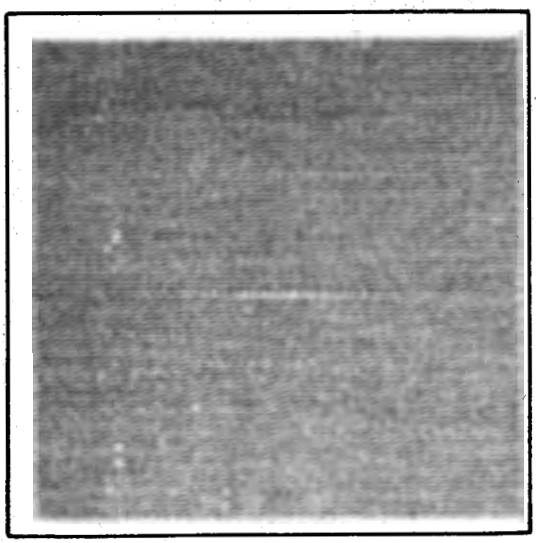
Motor:

Cotton piece goods, not otherwise indexed:
Motor volume minimum weight—any quantity.

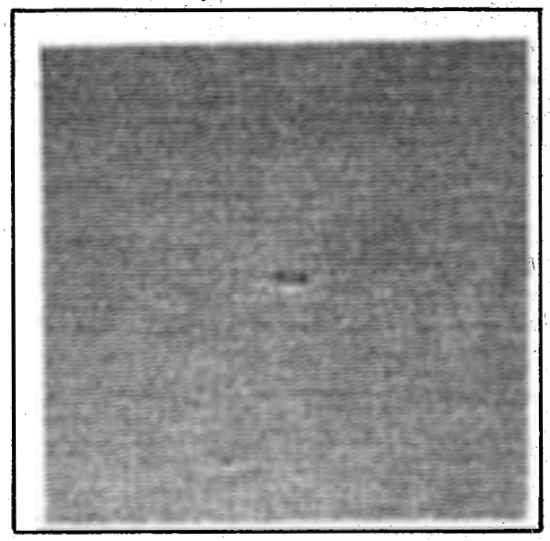
MILITARY INTEREST:
Army—GL
Navy—S&A
Air Force—ASD

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Slub



Slub (Loose, Thick)



Lump or Knot (Fuzz)

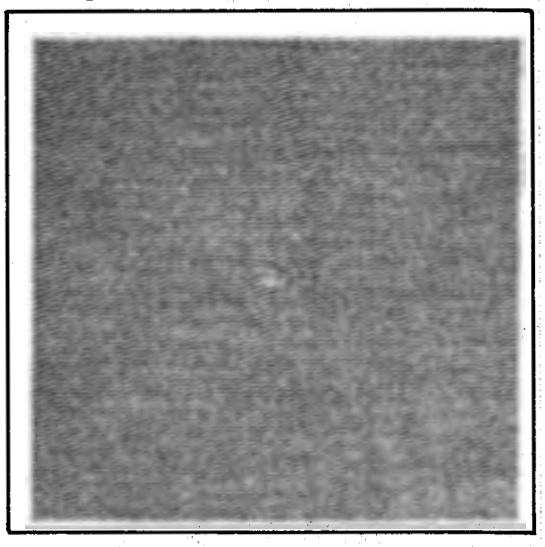


Figure 1 (Limits of Acceptability)

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